SE601P

(Software construction and development Practical)

Objectives from Week 01 – 08

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SE601P – Week 1 MCQs

Object-Oriented Software Development

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1. Object-oriented software development mainly focuses on:

- A) Functions and procedures
- B) Algorithms only
- ✔ C) Objects and their interactions
- D) Hardware components

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2. An object in object-oriented programming is best described as:

- A) A function
- ✔ B) An instance of a class
- C) A package
- D) A variable

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3. A class in object-oriented programming is:

- ✔ A) A blueprint for creating objects
- B) A specific instance
- C) A method
- D) A data type only

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4. Which of the following defines both data and behavior of an object?

- A) Interface
- B) Package
- ✔ C) Class
- D) Variable

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5. Which OOP principle hides data and allows access through methods only?

- ✔ A) Encapsulation
- B) Inheritance
- C) Polymorphism
- D) Abstraction

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6. Encapsulation helps in:

- A) Increasing code duplication
- ✔ B) Improving security and maintainability
- C) Reducing memory usage only
- D) Removing classes

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7. Inheritance is mainly used to:

- A) Hide implementation
- ✔ B) Reuse existing code
- C) Increase object lifetime
- D) Destroy objects

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8. Which relationship allows a class to acquire properties of another class?

- A) Association
- ✔ B) Inheritance
- C) Aggregation
- D) Composition

---

9. Polymorphism allows objects to:

- A) Have only one form
- ✔ B) Behave differently using the same interface
- C) Hide data
- D) Exist independently

---

10. Which concept allows the same method name to perform different actions?

- A) Encapsulation
- B) Inheritance
- ✔ C) Polymorphism
- D) Aggregation

---

11. Abstraction focuses on:

- ✔ A) Hiding unnecessary details
- B) Code duplication
- C) Memory allocation
- D) Object destruction

---

12. Which of the following is commonly used to achieve abstraction?

- A) Variables
- ✔ B) Interfaces and abstract classes
- C) Objects
- D) Constructors

---

13. Which term refers to characteristics of an object?

- ✔ A) Properties
- B) Methods
- C) Classes
- D) Packages

---

14. Methods in an object define:

- A) Object structure
- ✔ B) Object behavior
- C) Object memory size
- D) Object lifetime

---

15. The time period during which an object exists in memory is called:

- A) Object scope
- ✔ B) Object lifetime
- C) Object abstraction
- D) Object inheritance

---

16. An object is created when:

- ✔ A) Memory is allocated and properties are initialized
- B) Program ends
- C) Class is deleted
- D) Method is called

---

17. Object destruction occurs when:

- A) Object is accessed
- B) Method is executed
- ✔ C) Memory allocated to the object is released
- D) Class is inherited

---

18. Information hiding mainly aims to:

- ✔ A) Reduce dependency between program modules
- B) Increase execution speed
- C) Increase memory usage
- D) Remove methods

---

19. Which concept is closely related to information hiding?

- ✔ A) Encapsulation
- B) Polymorphism
- C) Inheritance
- D) Aggregation

---

20. Association in OOP represents:

- ✔ A) A relationship between independent objects
- B) A strong ownership
- C) Code reuse
- D) Method overloading

---

21. Which association allows both objects to access each other?

- A) Directional
- ✔ B) Bidirectional
- C) Single
- D) Abstract

---

22. In an e-commerce system, the relationship between Customer and Order is an example of:

- ✔ A) Association
- B) Composition
- C) Inheritance
- D) Polymorphism

---

23. Composition is a relationship where:

- ✔ A) Child object cannot exist without parent
- B) Objects exist independently
- C) Objects share memory
- D) Objects inherit behavior

---

24. Which example best represents composition?

- A) Teacher and Department
- B) Student and Course
- ✔ C) Car and Engine
- D) Customer and Order

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25. In composition, composed objects are:

- A) Shared among multiple objects
- ✔ B) Created and destroyed with the main object
- C) Independent of main object
- D) Stored globally

---

26. Aggregation differs from composition because in aggregation:

- ✔ A) Child objects can exist independently
- B) Child objects are destroyed with parent
- C) Ownership is strong
- D) Memory is shared

---

27. Which relationship shows weak ownership?

- A) Composition
- ✔ B) Aggregation
- C) Inheritance
- D) Encapsulation

---

28. School and Department relationship is an example of:

- A) Composition
- ✔ B) Aggregation
- C) Inheritance
- D) Polymorphism

---

29. Which relationship allows objects to be reused easily?

- A) Composition
- ✔ B) Aggregation
- C) Encapsulation
- D) Abstraction

---

30. Which of the following best differentiates aggregation from composition?

- A) Method overloading
- ✔ B) Object independence
- C) Data hiding
- D) Code reuse

SE601P – Week 2 MCQs

Information Sharing, Generalization, Realization & Polymorphism

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1. In object-oriented programming, information sharing between objects is mainly achieved through:

- A) Variables only
- ✔ B) Method calls and message passing
- C) Inheritance only
- D) Constructors

---

2. Which technique allows one object to invoke a method of another object?

- ✔ A) Method call
- B) Message passing
- C) Aggregation
- D) Generalization

---

3. Passing an object reference to a method is an example of:

- A) Inheritance
- ✔ B) Parameter passing
- C) Overriding
- D) Abstraction

---

4. In message passing, communication between objects occurs through:

- A) Direct data access
- ✔ B) Sending messages

- C) Global variables
- D) Static methods

---

5. Which of the following improves interaction while keeping objects loosely coupled?

- A) Direct data access
- ✔ B) Message passing
- C) Global variables
- D) Hard coding

---

6. Generalization in OOP is mainly used to:

- A) Hide data
- ✔ B) Create class hierarchies
- C) Destroy objects
- D) Reduce memory usage

---

7. In generalization, the existing class is called:

- A) Subclass
- ✔ B) Superclass
- C) Interface
- D) Object

---

8. A subclass in generalization:

- ✔ A) Inherits properties and methods of superclass
- B) Cannot add new features
- C) Cannot override methods
- D) Exists independently

---

9. Generalization helps in software development by:

- A) Increasing code duplication

- ✔ B) Reducing development and maintenance effort
- C) Reducing security
- D) Eliminating classes

---

10. Which mechanism allows a new class to be based on an existing class?

- A) Aggregation
- ✔ B) Inheritance
- C) Association
- D) Encapsulation

---

11. In inheritance, the derived class is also known as:

- A) Parent class
- ✔ B) Subclass
- C) Abstract class
- D) Interface

---

12. Adding new methods in a subclass that are not in the superclass is called:

- A) Overriding
- ✔ B) Extension
- C) Polymorphism
- D) Abstraction

---

13. Redefining an inherited method in a subclass is called:

- A) Extension
- ✔ B) Overriding
- C) Aggregation
- D) Generalization

---

14. Which of the following best describes method overriding?

- A) Same method name with different parameters

- ✔ B) Same method name with different implementation
- C) Different method name with same behavior
- D) Same method name in different classes only

---

15. In the Animal–Dog example, bark() method represents:

- A) Inheritance
- ✔ B) Extension
- C) Overriding
- D) Abstraction

---

16. Realization in OOP refers to:

- A) Extending a class
- ✔ B) Implementing an interface
- C) Aggregating objects
- D) Creating objects

---

17. An interface in OOP contains:

- ✔ A) Abstract method declarations
- B) Only variables
- C) Complete method implementations
- D) Constructors

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18. A class that realizes an interface must:

- A) Override superclass methods
- ✔ B) Implement all interface methods
- C) Extend another class
- D) Hide interface methods

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19. Realization helps in:

- A) Tight coupling

- ✔ B) Flexible and modular design
- C) Increasing dependency
- D) Eliminating polymorphism

---

20. Which keyword is commonly used to realize an interface in Java?

- A) extends
- ✔ B) implements
- C) inherits
- D) override

---

21. Polymorphism allows objects to:

- ✔ A) Take multiple forms
- B) Share memory
- C) Hide data
- D) Exist independently

---

22. Method overloading means:

- ✔ A) Same method name with different parameters
- B) Same method name with same parameters
- C) Different method name with same parameters
- D) Removing inherited methods

---

23. Method overriding occurs when:

- A) Method is overloaded
- ✔ B) Subclass changes inherited method behavior
- C) Interface is implemented
- D) Object is destroyed

---

24. Which type of polymorphism is achieved at runtime?

- A) Method overloading

- ✔ B) Method overriding
- C) Constructor overloading
- D) Operator overloading

---

25. Interfaces mainly support which type of polymorphism?

- A) Compile-time
- ✔ B) Runtime
- C) Static
- D) None

---

26. In the Vehicle–Car–Plane example, which concept is demonstrated?

- A) Encapsulation
- ✔ B) Polymorphism
- C) Aggregation
- D) Composition

---

27. The statement `Vehicle v = new Car();` demonstrates:

- ✔ A) Polymorphism
- B) Abstraction only
- C) Encapsulation
- D) Aggregation

---

28. Calling `v.move()` in polymorphism results in:

- A) Compile-time error
- ✔ B) Method of actual object being executed
- C) Base class method always
- D) Interface method

---

29. Which concept allows sharing behavior without sharing implementation?

- A) Inheritance

- ✔ B) Realization
- C) Aggregation
- D) Association

---

30. Which of the following improves code flexibility the most?

- A) Global variables
- ✔ B) Polymorphism
- C) Tight coupling
- D) Hard coding

Week 3 – MCQs

1. Generic programming allows you to:
A) Write code only for integers
B) Write reusable code for different data types ✔
C) Avoid using classes
D) Write code only for strings

2. In Java, which symbol is used for type parameters in generics?
A) []
B) <> ✔
C) {}
D) ()

3. What is the purpose of using constants in programming?
A) To store temporary values
B) To define values that do not change ✔
C) To confuse the programmer
D) To replace variables in loops

4. How should constants be named according to good programming practices?
A) lowercase letters with underscores
B) camelCase
C) All uppercase letters with underscores ✔
D) Random letters

5. Which of the following is a good programming practice?
A) Using meaningless variable names
B) Writing long functions with multiple tasks

C) Commenting complex logic ✔
D) Ignoring testing of code

6. In the object model case study, which object represents the link between a student and a course?
A) Student
B) Course
C) Enrollment ✔
D) Teacher

7. Distributed objects communicate with each other using:
A) Direct memory access
B) Remote procedure calls (RPCs) ✔
C) File transfer only
D) Local variables

8. Which of the following is NOT a technology used for distributed objects?
A) Java RMI
B) CORBA
C) .NET Remoting
D) HTML ✔

9. Object serialization converts an object into:
A) A string
B) A byte stream ✔
C) A method
D) A class

10. Deserialization is:
A) Converting byte stream back into an object ✔
B) Sending an object to the printer
C) Deleting an object from memory
D) Creating a new class

11. Which of the following formats can be used for object serialization?
A) JSON ✔
B) XML ✔
C) Protocol Buffers ✔
D) PDF

12. A peer-to-peer network use case of object serialization is:
A) Sending emails only
B) File sharing between peers ✔
C) Printing documents
D) Compiling code

13. Benefits of generic programming include:
A) Code duplication
B) Flexibility and reusability ✔
C) Slower execution
D) Only works with one type of data

14. Which of the following is NOT a benefit of distributed objects?
A) Scalability
B) Fault-tolerance
C) Centralized single machine requirement ✔
D) Modular architecture

15. Where should constants be placed for easy access?
A) At the end of the code
B) Inside every function
C) In a central location ✔
D) Randomly in code

Week 4 – MCQs

1. What is exception handling in programming?
A) Ignoring runtime errors
B) Detecting and responding to unexpected errors ✔
C) Manually stopping program execution
D) Writing faster code

2. Which block always executes whether an exception occurs or not?
A) try
B) catch
C) finally ✔
D) throw

3. Which of the following is a system exception in Java?
A) FileNotFoundException ✔
B) WithdrawalException
C) NegativeNumberException
D) CustomException

4. Which of the following is a user-defined exception?
A) ArithmeticException
B) FileNotFoundException

C) NegativeNumberException ✔
D) IOException

5. What keyword is used to throw an exception in Java?
A) throw ✔
B) throws
C) catch
D) try

6. How do you catch multiple types of exceptions?
A) Using multiple try blocks
B) Using multiple catch blocks ✔
C) Using finally block
D) Using throw keyword

7. What is the difference between exception handling and error handling?
A) Exception handling is for serious system errors
B) Error handling is for recoverable program errors
C) Exception handling handles recoverable errors ✔
D) Both are the same

8. Which of the following is an example of a business exception?
A) NullPointerException
B) OutOfMemoryError
C) WithdrawalException ✔
D) FileNotFoundException

9. In the example `int result = 10 / 0;`, which exception is thrown?
A) IOException
B) ArithmeticException ✔
C) NegativeNumberException
D) FileNotFoundException

10. Why do we use user-defined exceptions?
A) To handle system errors only
B) To implement custom business logic errors ✔
C) To avoid using try-catch blocks
D) To make code slower

11. What will happen if an exception is not caught in Java?
A) Program continues normally
B) Program crashes and prints stack trace ✔
C) Exception is ignored
D) Program becomes faster

12. Which of the following keywords is used to declare that a method may throw an exception?
A) try
B) throw
C) throws ✔
D) catch

13. What is the purpose of a try block?
A) To define a method
B) To handle exceptions
C) To enclose code that may throw an exception ✔
D) To terminate program

14. Which of these is NOT an exception in Java?
A) FileNotFoundException
B) NegativeNumberException
C) OutOfMemoryError ✔
D) IOException

15. In user-defined exceptions, which class is extended to create a custom exception?
A) RuntimeException
B) Exception ✔
C) Error
D) Object

Week 5 – MCQs

1. What is an Integrated Development Environment (IDE)?
A) A hardware device
B) A software application providing a complete development environment ✔
C) A programming language
D) A database system

2. Which of the following is a popular IDE used mainly for Java development?
A) Git
B) Eclipse ✔
C) Jenkins
D) Gradle

3. Visual Studio Code is best described as:
A) A database tool
B) A lightweight code editor with extensions ✔
C) A version control system
D) A CI/CD tool

4. Version Control Systems (VCS) are used to:
A) Compile source code
B) Design user interfaces
C) Manage changes to source code over time ✔
D) Test software automatically

5. Which of the following is a Version Control System?
A) Maven
B) Git ✔
C) Jenkins
D) Android Studio

6. Git Bash is mainly used for:
A) Writing Java code
B) Running Git commands on Windows ✔
C) Designing mobile apps
D) Managing databases

7. Which command-line environment does Git Bash provide?
A) Windows-only commands
B) Linux-like terminal ✔
C) Android emulator
D) Cloud environment

8. Build automation tools are used to:
A) Write source code
B) Design UML diagrams
C) Automate building and packaging of software ✔
D) Store project documentation

9. Which of the following is a build automation tool?
A) SVN
B) Gradle ✔
C) Git Bash
D) CircleCI

10. Which build tool is commonly used with Android Studio?
A) Ant
B) Maven
C) Gradle ✔
D) Jenkins

11. Continuous Integration (CI) mainly focuses on:
A) Manual testing
B) Automatic testing after code changes ✔

C) Writing documentation
D) Designing software architecture

12. Continuous Delivery (CD) refers to:
A) Writing unit tests
B) Managing source code
C) Automatically deploying tested software ✔
D) Debugging code manually

13. Which of the following is a CI/CD tool?
A) IntelliJ
B) CircleCI ✔
C) Git Bash
D) Eclipse

14. Jenkins is primarily used for:
A) Editing code
B) Version control
C) Automating build, test, and deployment ✔
D) Database management

15. Which tool helps developers collaborate by tracking code changes?
A) Maven
B) Visual Studio
C) Git ✔
D) Android Studio

Week 6 – MCQs

- **Code review is mainly used to**

A) Increase program execution speed
B) Identify bugs and improve code quality ✔
C) Replace testing phase
D) Reduce hardware cost

- **Code review is performed by**

A) End users
B) Project manager only
C) Developers reviewing each other's code ✔
D) System analysts

- **Which activity helps reduce technical debt?**

A) Debugging
B) Code compilation
C) Code review ✔
D) Deployment

- **Code review helps in**

A) Increasing file size
B) Improving maintainability ✔
C) Removing documentation
D) Avoiding version control

- **Which platform provides built-in pull request reviews?**

A) Jira
B) GitHub ✔
C) Selenium
D) JMeter

- **Bitbucket integrates closely with**

A) Jenkins
B) Jira ✔
C) Git Bash
D) Eclipse

- **Which tool provides code quality and technical debt analysis?**

A) Review Board
B) CodeClimate ✔
C) GitLab
D) Crucible

- **GitLab provides**

A) Only bug tracking
B) Code review only
C) Code review with CI/CD features ✔
D) Testing automation only

- **Crucible is developed by**

A) Microsoft
B) Google
C) Atlassian ✔
D) Oracle

- **Review Board supports which version control systems?**

A) Git only
B) SVN only

C) Git, Mercurial, and Subversion ✔
D) CVS only

- **Bug tracking tools are used to**

A) Write code
B) Design UI
C) Track bugs and defects ✔
D) Compile programs

- **Bug tracking tools help teams to**

A) Increase bugs
B) Delay development
C) Fix issues early ✔
D) Avoid documentation

- **Which feature is NOT part of bug tracking tools?**

A) Issue prioritization
B) Status tracking
C) Code compilation ✔
D) Issue assignment

- Jira is mainly used as a

A) Testing tool
B) Bug tracking tool ✔
C) IDE
D) Compiler

- **Bugzilla is an example of**

A) Documentation tool
B) Bug tracking tool ✔
C) Build tool
D) Testing framework

- **Software documentation tools are used to**

A) Test applications
B) Track bugs
C) Create and manage documentation ✔
D) Compile code

- **Documentation tools improve**

A) Hardware performance
B) Team communication ✔
C) Code execution speed
D) Memory usage

- **Which is a software documentation tool?**

A) Selenium
B) JMeter
C) Confluence ✔
D) Git

- **Read the Docs is mainly used for**

A) Bug tracking
B) Software documentation ✔
C) Code compilation
D) Version control

- **Sphinx is used to**

A) Test APIs
B) Create documentation ✔
C) Track issues
D) Deploy software

- **Testing tools are used to**

A) Write documentation
B) Automate testing ✔
C) Review code
D) Track bugs

- **Testing tools help reduce**

A) User interface
B) Manual testing effort ✔
C) Code size
D) Documentation

- **Selenium is mainly used for**

A) Performance testing
B) API testing
C) Automated browser testing ✔
D) Bug tracking

- **JMeter is commonly used for**

A) Unit testing
B) Performance testing ✔
C) UI design
D) Code review

- **Postman is mainly used for**

A) Web testing
B) API testing ✔

C) Code documentation
D) Bug tracking

- **SoapUI is used for**

A) API testing ✔
B) Code compilation
C) Version control
D) Documentation

- **TestRail is used for**

A) Source control
B) Test case management ✔
C) Code review
D) Deployment

- **Testing tools help ensure**

A) Faster typing
B) Compliance with standards ✔
C) Larger codebase
D) Less documentation

- **Bug tracking tools are most useful for**

A) Small scripts
B) Large and complex projects ✔
C) Single-user programs
D) Offline systems

- **Code review tools mainly improve**

A) Internet speed
B) Team collaboration ✔
C) Hardware usage
D) Screen resolution

Week 7 – MCQs

- **Jira is mainly used for**

A) Code compilation
B) Project management and issue tracking ✔
C) UI designing
D) Database management

- **Before using Jira, a user must**

A) Install Java
B) Create a Jira account ✔
C) Write source code
D) Create a repository

- **In Jira, a project is used to**

A) Store documentation only
B) Contain issues and tasks ✔
C) Compile software
D) Test applications

- **Creating a new project in Jira allows users to**

A) Delete issues
B) Organize work for a specific application ✔
C) Install plugins
D) Design databases

- **Project setup in Jira includes**

A) Writing code
B) Defining workflows and permissions ✔
C) Compiling files
D) Running test cases

- **Issues in Jira represent**

A) Hardware problems
B) Tasks, bugs, or user stories ✔
C) Only documentation
D) Network errors

- **Which of the following is NOT an issue type in Jira?**

A) Bug
B) Task
C) Epic
D) Compiler ✔

- **Adding issues in Jira helps to**

A) Increase bugs
B) Track work progress ✔
C) Reduce storage
D) Disable collaboration

- **Jira can be connected to which tools?**

A) Text editors
B) Source Code Management tools ✔

C) Media players
D) Browsers

- **GitHub integration with Jira allows**

A) UI designing
B) Linking code changes with issues ✔
C) Database backup
D) Software installation

- **An Epic in Jira is**

A) A small task
B) A large body of work ✔
C) A testing report
D) A bug only

- **Epics are mainly used for**

A) Code execution
B) High-level planning ✔
C) Memory management
D) Compilation

- **Child issues are**

A) Independent projects
B) Smaller tasks under an Epic ✔
C) System errors
D) User accounts

- **Setting timelines in Jira helps to**

A) Delay project
B) Plan start and end dates ✔
C) Increase cost
D) Remove issues

- **Roadmap in Jira provides**

A) Code editor
B) Visual overview of project progress ✔
C) Compiler output
D) Error logs

- **Monitoring progress in Jira helps teams to**

A) Ignore deadlines
B) Track completed and pending work ✔
C) Increase errors
D) Remove documentation

- **Roadmap is mainly used by**

A) End users
B) Project managers and teams ✔
C) Hardware engineers
D) Network admins

- **Jira issues can be**

A) Assigned to team members ✔
B) Compiled
C) Executed
D) Installed

- **Connecting Jira with SCM tools improves**

A) Hardware speed
B) Traceability between code and tasks ✔
C) Screen resolution
D) File compression

- **Jira is commonly used in**

A) Graphic design
B) Software development projects ✔
C) Video editing
D) Hardware manufacturing

Week 8 – MCQs

1. What is the first step in creating a new project in Asana?
A) Add tasks
B) Create a new project ✔
C) Assign team members
D) Set timelines

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2. In Asana, who can assign tasks to team members?
A) Only developers
B) Project manager ✔
C) Any team member without restrictions
D) Only the admin

---

3. What feature in Asana allows tracking of task completion and progress?
A) Work Space
B) Project Manager Dashboard ✔
C) Developer View
D) Team Settings

---

4. What is the purpose of setting timelines for a task in Asana?
A) To decorate the project
B) To determine start and end dates for task completion ✔
C) To add comments
D) To assign priority only

---

5. In Asana, what view shows pending and completed tasks for a developer?
A) Dashboard view
B) Epic view
C) Developer View ✔
D) RoadMap view

---

6. What is a Work Space in Asana?
A) A tool for coding
B) A central area to organize projects and teams ✔
C) A timeline chart
D) A type of report

---

7. How do you add tasks in Asana?
A) By creating a new project
B) By clicking "Add Task" within the project ✔
C) By exporting tasks from another tool
D) Tasks are automatically added

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8. What is the purpose of a Project Manager Dashboard in Asana?
A) To write code
B) To monitor project progress and deadlines ✔
C) To assign tasks automatically
D) To generate test cases

---

9. What can you do in the Developer View in Asana?

A) Track project budget

B) View pending and completed tasks ✔

C) Assign tasks to others

D) Create new projects

---

10. What is the main purpose of adding a timeline to a task?

A) For decoration

B) To track the duration and deadlines ✔

C) To set priorities only

D) To assign to multiple teams